

# Giacomo Galloni

Postdoctoral Researcher — Cosmology · University of Ferrara

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## RESEARCH INTERESTS

My research focuses on cosmic inflation through the Cosmic Microwave Background (CMB) and Gravitational Waves (GWs), with a methodological emphasis on likelihood analysis. I lead the large-scale CMB likelihood effort within the LiteBIRD collaboration and co-lead the Likelihood & Theory group of the Simons Observatory (SO), where I develop end-to-end pipelines that combine harmonic and pixel-based estimators with realistic component separation. In parallel, I am building a unified likelihood framework intended for use across current and future CMB experiments. On the science side, I have produced Bayesian and frequentist constraints on the amplitude and tilt of the primordial tensor spectrum, and I work on stochastic GW backgrounds and their cross-correlation with the CMB, including recent data-driven bounds on the quantum decoherence of inflationary GWs.

## POSITIONS & EDUCATION

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| Feb 2024 — present  | <b>Postdoctoral Researcher</b> , University of Ferrara.                                                                                                                |
| Nov 2020 — Jan 2024 | <b>Ph.D. with honors in Astronomy, Astrophysics and Space Science</b> , University of Roma Tor Vergata.<br>Supervisors: Prof. Nicola Vittorio, Prof. Sabino Matarrese. |
| Oct 2018 — Oct 2020 | <b>M.Sc. with honors, Physics of the Universe</b> , University of Padova. Thesis: Prof. Nicola Bartolo, Dr. Angelo Ricciardone.                                        |
| Oct 2014 — Oct 2018 | <b>B.Sc. in Physics</b> , University of Padova. Thesis: Prof. Giacomo Ciani.                                                                                           |

## FELLOWSHIPS, VISITINGS & SCHOOLS

|                     |                                                                    |          |                                                                            |
|---------------------|--------------------------------------------------------------------|----------|----------------------------------------------------------------------------|
| Sep — Dec 2023      | Funded visiting at IJCLab, Orsay (France).                         | Mar 2022 | GGI workshop on Astroparticle Physics, Cosmology and Gravitation, Firenze. |
| Jun — Jul 2023      | Funded visiting at Kavli IPMU, Tokyo (Japan) via MSCA CMB-INFLATE. | Dec 2021 | Tonale Winter School in Cosmology, University of Heidelberg.               |
| Nov 2020 — Jan 2024 | Full Ph.D. fellowship, University of Roma La Sapienza.             | Nov 2021 | Workshop on CLASS and Cobaya, University of Padova.                        |
| Jun — Sep 2017      | Funded internship at the University of Florida (bachelor thesis).  |          |                                                                            |

## SELECTED PUBLICATIONS

Full list on [Google Scholar](#) and [INSPIRE-HEP](#). First-author and small-group papers below.

- Genesini, V., G. Galloni, et al. (2026). “Cross-spectra likelihood for robust  $\tau$  constraints from all satellite polarisation data”. arXiv:2603.22454 [astro-ph.CO]
- Galloni, G., P. Campeti, et al. (2025). “Accurate and efficient likelihood modeling for large-scale CMB data”. JCAP 12, p. 052 — doi:10.1088/1475-7516/2025/12/052 — arXiv:2505.24829
- Giardiello, S., A. J. Duivenvoorden, et al. (2025). “Modeling beam chromaticity for high-resolution CMB analyses”. Phys. Rev. D 111, p. 043502 — arXiv:2411.10124
- Kruijf, J. de, G. Galloni, et al. (2025). “The first data-driven bounds on the quantum decoherence of inflationary gravitational waves”. arXiv: 2511.14727 [astro-ph.CO]
- Galloni, G., S. Henrot-Versillé, et al. (2024). “Robust constraints on tensor perturbations from cosmological data: a Bayesian and frequentist comparative analysis”. Phys. Rev. D 110.6, p. 063511 — arXiv:2405.04455
- Galloni, G., M. Ballardini, et al. (2023). “Unraveling the CMB lack-of-correlation anomaly with the cosmological GW background”. JCAP 10, p. 013 — arXiv:2305.18184
- Galloni, G., N. Bartolo, et al. (2023). “Updated constraints on amplitude and tilt of the tensor primordial spectrum”. JCAP 04, p. 062 — arXiv:2208.00188
- Galloni, G., N. Bartolo, et al. (2022). “Test of the statistical isotropy of the universe using gravitational waves”. JCAP 09, p. 046 — arXiv:2202.12858

## INDEPENDENT PROJECTS

|         |                                                                                                                                                                                                                                   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ongoing | <b>CosmoForge — Power Spectrum and Likelihood Framework.</b> Python framework for power-spectrum estimation and likelihood analysis on large-scale CMB data, designed as a unified tool for LiteBIRD, SO, and future experiments. |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**BICEP Diagnosis with the marginalized HL Likelihood.** Re-analysis of BICEP data exploiting new capabilities of the marginalized Hamimeche & Lewis (mHL) likelihood, stress-testing existing constraints on tensor modes.

## COLLABORATION PROJECTS

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|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Jun 2024</i> — | <b>LiteBIRD — Foreground and Sensitivity Studies.</b> Interfacing my Python package with foreground separation and sensitivity analyses.                                          |
| <i>May 2024</i> — | <b>SO — Likelihood &amp; Theory.</b> Co-lead of the group; maintenance and development of SOLikeT. Contributed single-mode likelihoods and led the restructuring of LAT MFLike.   |
| <i>Sep 2024</i> — | <b>SO — Power Spectrum.</b> Contributing to the Power Spectrum group, with planned developments on QML-based estimators for large-scale modes.                                    |
| <i>Jan 2026</i> — | <b>LiteBIRD — Moment-Expansion Component Separation.</b> Improving and characterizing the moment-expansion approach in the LiteBIRD context.                                      |
| <i>Jun 2023</i> — | <b>LiteBIRD — MaPLEs team.</b> Leading efforts to study and characterize the likelihood of LiteBIRD; released within the collaboration a Python package to perform this analysis. |
| <i>Apr 2023</i> — | <b>LiteBIRD — Simulation Team.</b> Validation tests, cosmological analysis of CMB temperature with LiteBIRD, and development of parts of <code>litebird_sim</code> .              |
| <i>May 2022</i> — | <b>LiteBIRD — Tests on Cosmic Inflation.</b> Monte-Carlo study of which inflationary models can be detected by LiteBIRD.                                                          |

## SERVICE & LEADERSHIP

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|-------------------|-----------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------|
| <i>Sep 2024</i> — | Co-lead, Likelihood & Theory group, Simons Observatory.   | <i>Jan 2026</i> | SOC member, LiteBIRD Face-to-Face Meeting.                                              |
| <i>Dec 2025</i> — | Early Career Coordination (ECC) representative, LiteBIRD. | <i>Jul 2024</i> | Session organizer & chair on the present/future of CMB, Marcel Grossman XVII (Pescara). |

## SELECTED TALKS

|                 |                                                                     |                 |                                                                   |
|-----------------|---------------------------------------------------------------------|-----------------|-------------------------------------------------------------------|
| <i>May 2026</i> | Invited Seminar at SISSA (Italy).                                   | <i>Jul 2024</i> | Invited Speaker, Marcel Grossman XVII, Pescara (Italy) — ICRANet. |
| <i>Apr 2026</i> | Invited Speaker, GRaviCon 2026, Pisa (Italy).                       | <i>Jul 2024</i> | Invited Speaker, KEK CMB Group Meeting, Tsukuba (Japan).          |
| <i>Nov 2025</i> | Invited Speaker, GW-CMB 2025, National Central University (Taiwan). | <i>Feb 2024</i> | Invited Speaker, ASI-LiteBIRD, Frascati (Italy).                  |
| <i>Sep 2025</i> | Invited Speaker, Tensions in Cosmology, Corfu (Greece).             | <i>Dec 2023</i> | Invited Speaker, Univers du pôle AzC, Orsay (France).             |
| <i>May 2025</i> | Invited Outreach Talk, Pint of Science, Ferrara (Italy).            | <i>May 2023</i> | Invited Seminar, University of Padova (Italy).                    |
| <i>Dec 2024</i> | Invited Seminar, University of Rome Tor Vergata (Italy).            | <i>Oct 2022</i> | Invited Speaker, LISA Community Call.                             |
| <i>Dec 2024</i> | Invited Seminar, University of Padova (Italy).                      | <i>Apr 2022</i> | Invited Seminar, University of Ferrara (Italy).                   |

## MENTORSHIP

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| <i>Feb 2026</i> —     | External mentor of a Ph.D. student at National Central University (Taiwan), working on $r-n_t$ constraints from a joint analysis of LIGO-Virgo-KAGRA likelihoods and Simons Observatory forecasts.                        |
| <i>Nov 2025</i> —     | Co-supervisor of a Master student at the University of Padova, on the recovery of $r$ and $n_t$ from SO forecasts.                                                                                                        |
| <i>Apr 2024</i> —     | Co-supervisor of a Ph.D. student at the University of Ferrara, who developed a cross-spectra likelihood for $\tau$ (Genesini et al. 2026), now working on moment-expansion component separation and reionization history. |
| <i>Feb — Nov 2024</i> | Supervisor of a Master student, Ferrara. <i>Thesis</i> : “Determination of the Reionization Optical Depth through a Combined Analysis of Planck and WMAP data.”                                                           |

## SOFTWARE SKILLS

**Statistics.** Expert in Bayesian and frequentist methods. Author of CosmoForge, a unified Python framework for QML and pixel-based CMB likelihood analyses; contributed to Cobaya; developed a profile-likelihood sampler.

**Programming.** Highly proficient in Python and C, with strong experience in package design, maintenance, and CI workflows. Author of CosmoForge, maintainer of SOLikeT, contributor to `litebird_sim` and several open-source cosmology repositories.

**Boltzmann codes.** Proficient user of CAMB and CLASS.